INDUSTRIAL INTERNSHIP REPORT ON

IOT-BASED SMART HELMET FOR MINER SAFETY AND REAL-TIME HAZARD MONITORING

Prepared by

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Executive Summary

This report presents the work completed during the Embedded Systems & IoT Industrial Internship conducted by upskill Campus in collaboration with The IoT Academy and UniConverge Technologies Pvt. Ltd. The internship provided practical exposure to embedded systems, IoT technologies, cloud connectivity, and real-time monitoring applications.

The project undertaken during the internship is titled "IoT-Based Smart Helmet for Miner Safety and Real-Time Hazard Monitoring." The objective of this project is to improve the safety of miners by continuously monitoring hazardous environmental conditions such as toxic gas leakage, fire, high temperature, and helmet removal or accidental fall.

The system is built using an ESP32 microcontroller integrated with an MQ135 gas sensor, DHT11 temperature sensor, IR sensor, fire sensor, OLED display, buzzer, and the Blynk IoT platform for remote monitoring and instant notifications.

This internship enhanced my understanding of embedded systems, sensor interfacing, IoT communication, cloud-based monitoring, and practical engineering problem-solving. It provided valuable hands-on learning and strengthened my technical skills for future career opportunities.

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1 PREFACE

This report presents the work carried out during my six-week **Embedded Systems & IoT Internship** conducted by **upskill Campus (USC)** in collaboration with **UniConverge Technologies (UCT)**. The internship provided me with an excellent opportunity to gain practical knowledge and understand how embedded systems and IoT technologies are applied to solve real-world industrial problems.

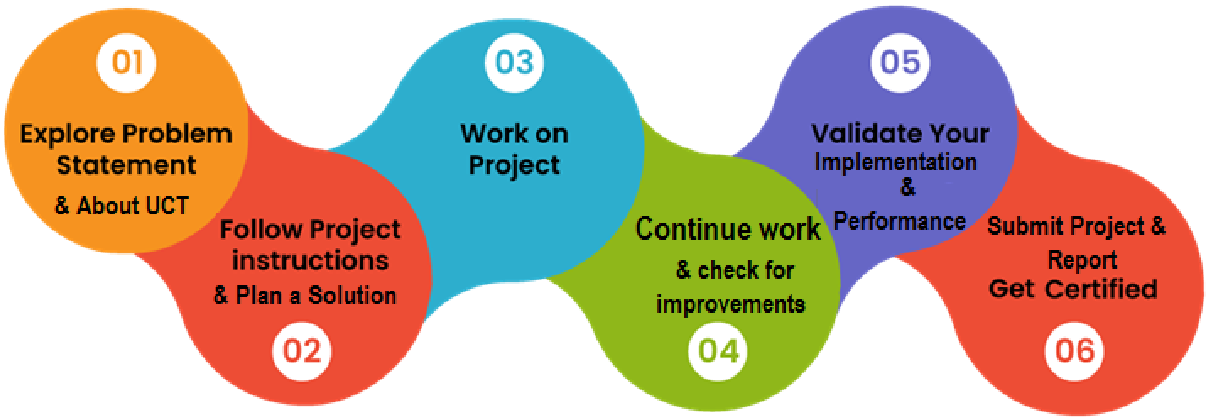
During the internship, I learned the fundamentals of embedded systems, sensor interfacing, IoT platforms, cloud connectivity, GitHub, project documentation, and real-time monitoring applications. The structured learning approach, assignments, and project work helped me strengthen both my technical and problem-solving skills.

As part of this internship, I completed the project **"IoT-Based Smart Helmet for Miner Safety and Real-Time Hazard Monitoring."** The project aims to improve the safety of miners by continuously monitoring hazardous gases, temperature, fire, and helmet status using an ESP32 microcontroller and multiple sensors. The system also provides real-time monitoring and alerts through the Blynk IoT platform.

This internship has enhanced my confidence, practical knowledge, and understanding of industry-oriented project development. The experience has motivated me to continue learning new technologies and prepare myself for future career opportunities in Embedded Systems and IoT.

I sincerely thank **upskill Campus (USC)**, **UniConverge Technologies (UCT)**, **The IoT Academy**, and everyone who supported and guided me throughout this internship. Their guidance and encouragement made this learning journey successful and memorable.

Finally, I encourage my juniors and peers to actively participate in internships and practical projects, as they provide valuable industry exposure, improve technical skills, and help build confidence for future career opportunities.



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2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

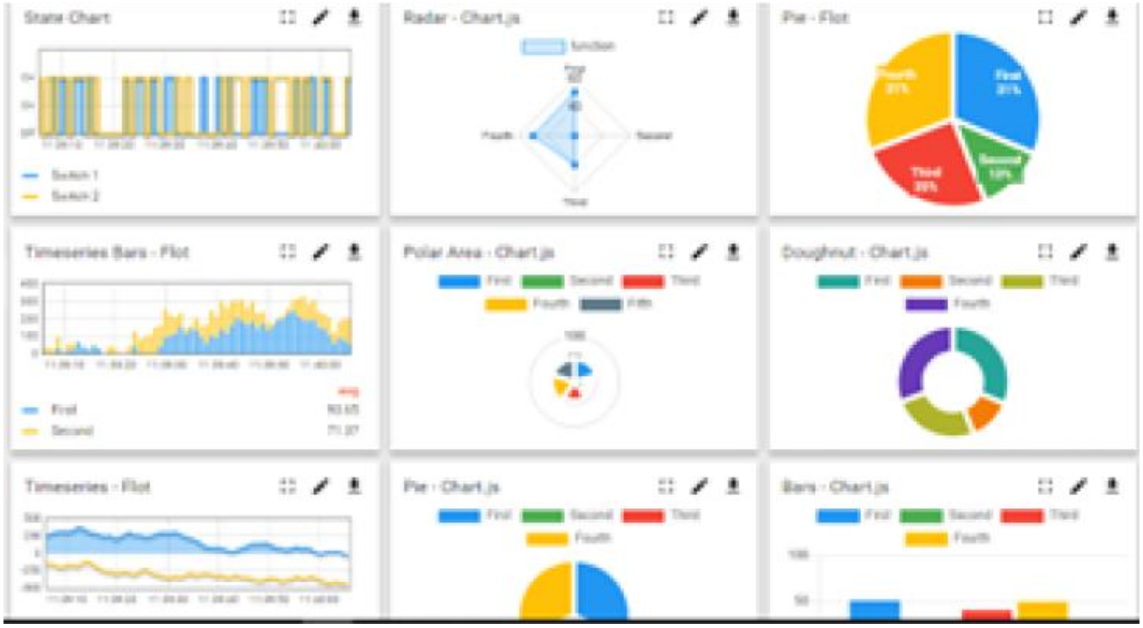
i. UCT IoT Platform (uct Insight)

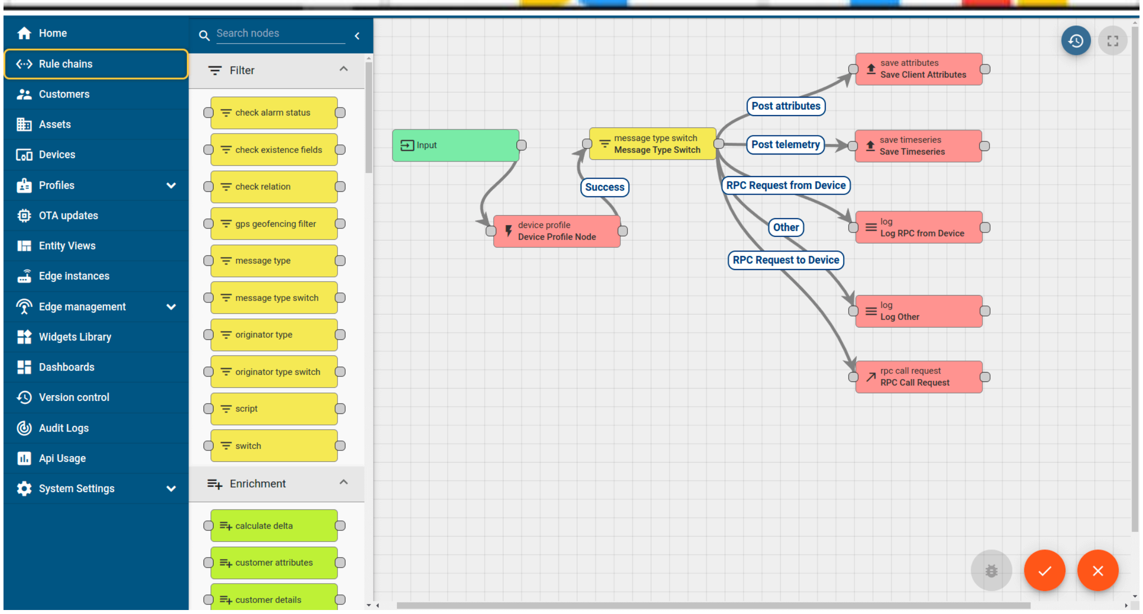
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

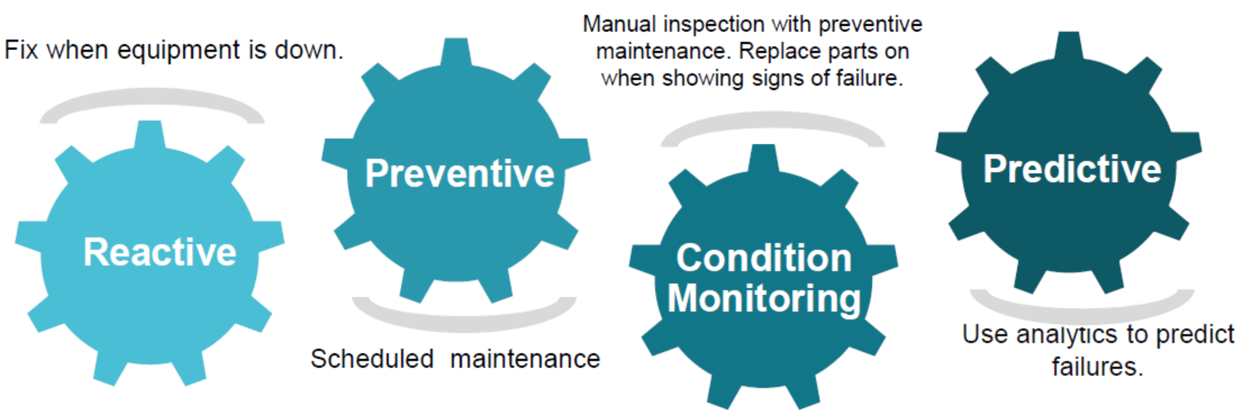
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

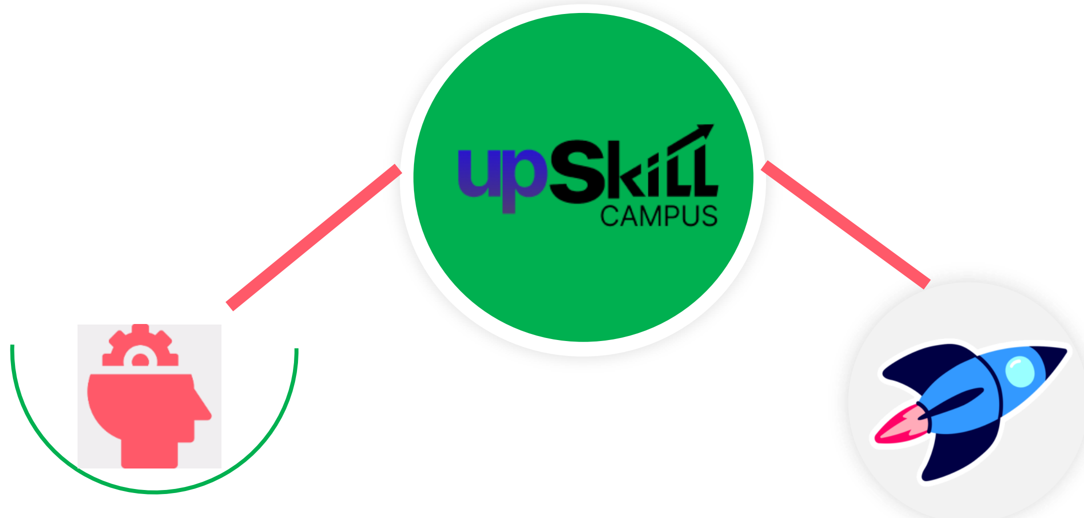
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

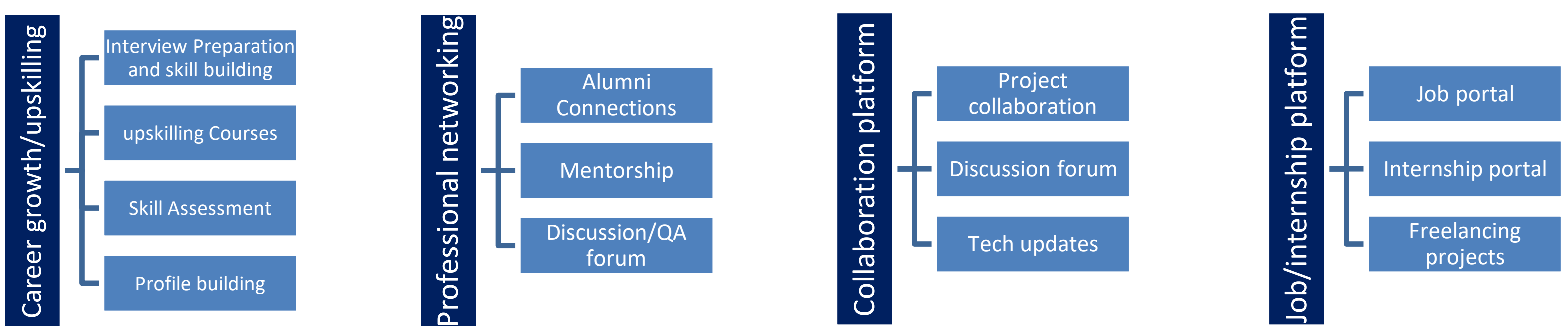
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1]Espressif Systems. *ESP32 Technical Reference Manual*. Espressif Systems, 2023. Available: https://www.espressif.com/sites/default/files/documentation/esp32_technical_reference_manual_en.pdf

[2] Blynk Inc. *Blynk IoT Platform Documentation*. Blynk, 2023. Available: https://docs.blynk.io

[3] Aosong Electronics. *DHT11 Humidity and Temperature Sensor Datasheet*. Aosong Electronics Co. Ltd., 2021.

2.6 Glossary

Terms	Acronym
IoT	Internet of Things
ESP32	Espressif 32-bit Microcontroller

MQ135	Metal Oxide Semiconductor Gas Sensor
DHT11	Digital Humidity and Temperature Sensor
IR Sensor	Infrared Sensor

3 Problem Statement

In the assigned problem statement

Mining is one of the most hazardous occupations due to the presence of toxic gases, high temperatures, fire hazards, and limited visibility inside underground mines. Traditional safety equipment does not provide continuous monitoring of environmental conditions, making it difficult to detect dangerous situations in real time.

The objective of this project is to develop an IoT-based smart helmet capable of monitoring hazardous gases, temperature, fire, and helmet status using multiple sensors connected to an ESP32 microcontroller. The system provides real-time monitoring through the Blynk IoT platform and generates instant alerts whenever unsafe conditions are detected, thereby improving the safety of mining workers.

4 Existing and Proposed solution

Existing Solution

Traditional mining safety helmets mainly provide protection against physical injuries but do not monitor environmental hazards such as toxic gas leakage, fire, or abnormal temperature. In most mining sites, environmental monitoring is performed using separate devices, which may delay hazard detection and emergency response. This increases the risk to workers operating in underground mines.

Proposed Solution

The proposed system is an IoT-based smart helmet that integrates an ESP32 microcontroller with an MQ135 gas sensor, DHT11 temperature sensor, IR sensor, and fire sensor. The system continuously monitors the surrounding environment and displays real-time information on an OLED display. Sensor data is transmitted to the Blynk IoT platform through Wi-Fi for remote monitoring. Whenever hazardous conditions such as toxic gas leakage, high temperature, fire, or helmet removal are detected, the system activates a buzzer and sends instant notifications to ensure timely action and improve miner safety.

Value Addition

The proposed IoT-based Smart Helmet enhances traditional mining safety by integrating multiple sensors into a single wearable device for continuous monitoring. It provides real-time detection of hazardous gases, high temperature, fire, and helmet removal, helping to prevent accidents before they become critical.

The integration of the ESP32 microcontroller with the Blynk IoT platform enables remote monitoring and instant alerts through a mobile application. The OLED display provides live sensor readings, while the buzzer offers immediate on-site warnings during emergencies.

Compared to conventional safety helmets, this system improves worker safety, reduces response time during hazardous situations, enables continuous environmental monitoring, and offers a cost-effective and scalable solution for mining industries.

4.1 Code submission (Github link)

https://github.com/ankishavishwakarma9-lgtm/upskillCampus/blob/main/Smart_Helmet_Code.ino

4.2 Report submission (Github link) :

5 Proposed Design/ Model

5.1 The proposed system is an IoT-based smart helmet designed to improve the safety of miners working in hazardous environments. The system uses an ESP32 microcontroller as the central processing unit to collect and process data from multiple sensors. The sensors continuously monitor environmental conditions such as toxic gas concentration, temperature, fire, and helmet status.

5.2 The processed data is displayed locally on an OLED display and simultaneously transmitted to the Blynk IoT platform through Wi-Fi. Whenever abnormal conditions are detected, the system activates a buzzer and sends instant notifications to the user through the Blynk mobile application. This enables real-time monitoring and rapid response during emergency situations.

5.3 High Level Diagram (if applicable)

remote supervision, enhancing the overall effectiveness of the smart helmet system.

3.2 Block Diagram

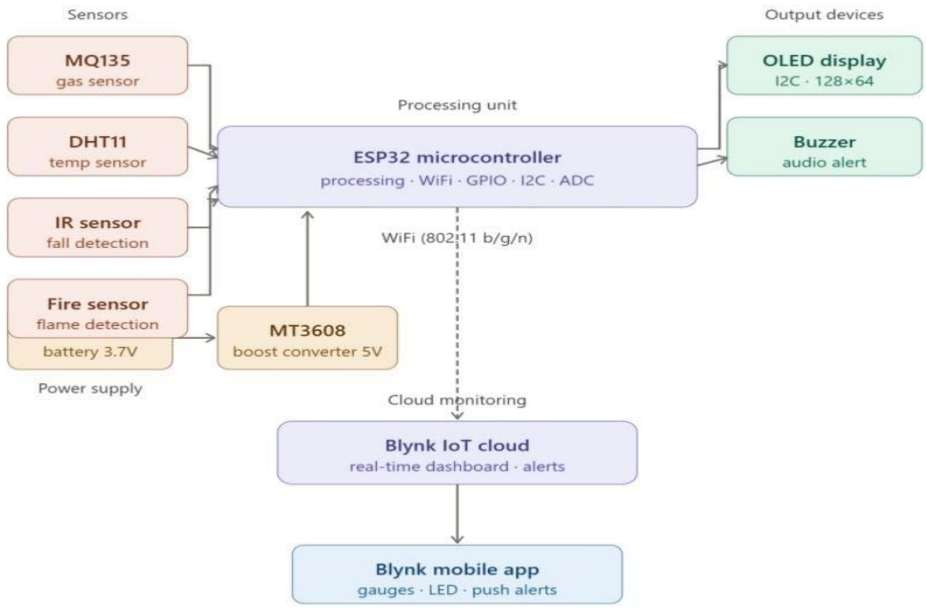
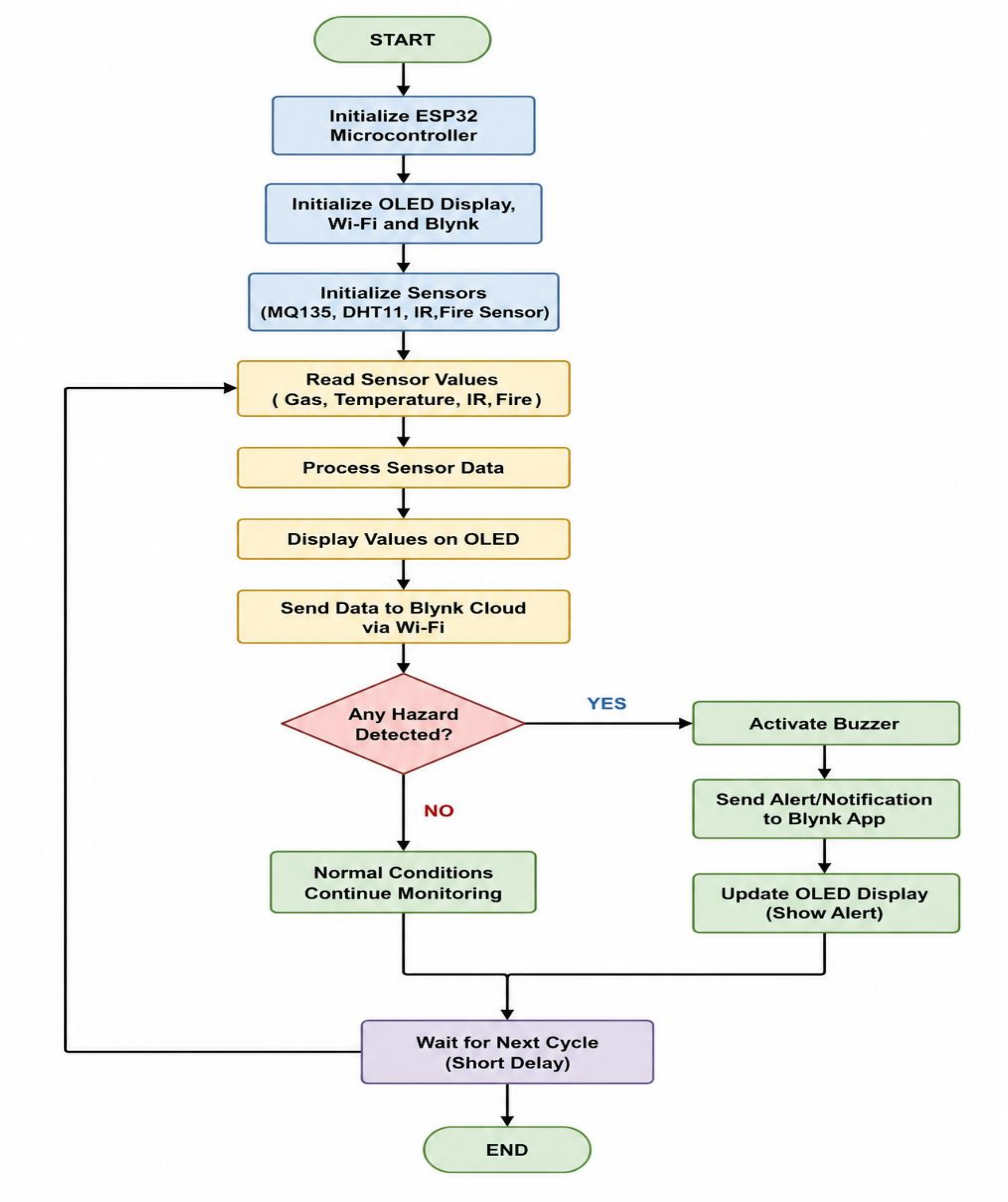


Figure 3.1 : Block Diagram of IoT Smart Helmet

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

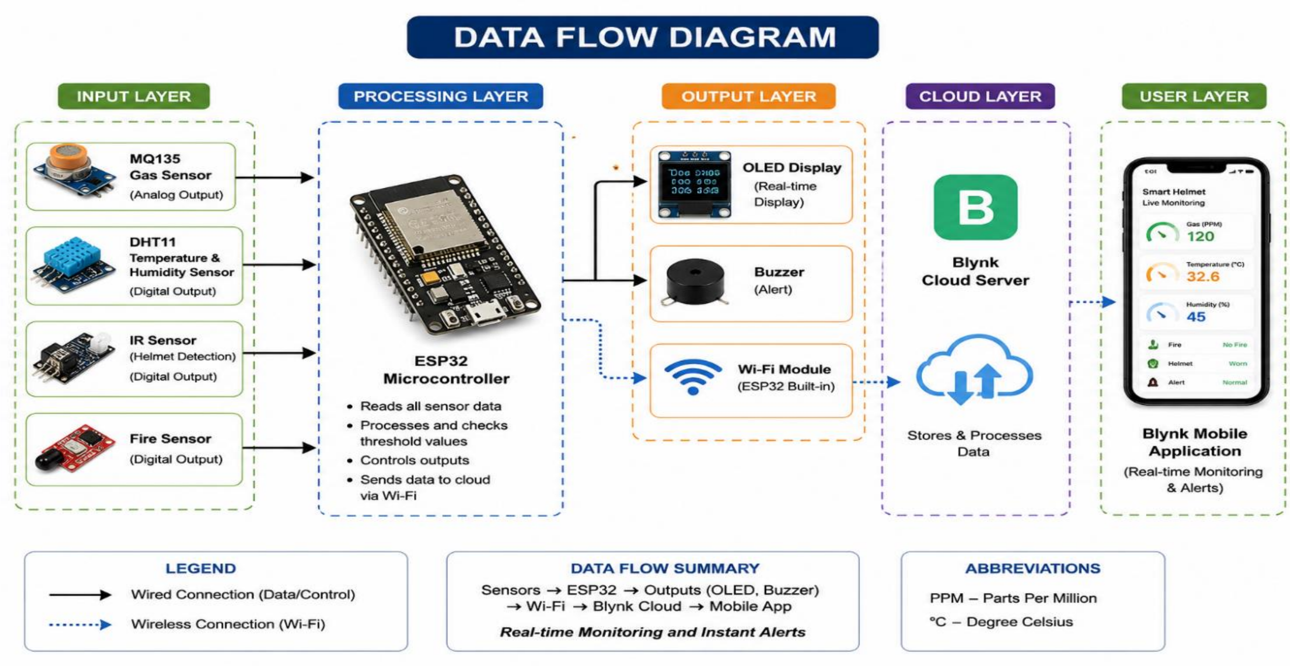
5.4 Low Level Diagram (if applicable)

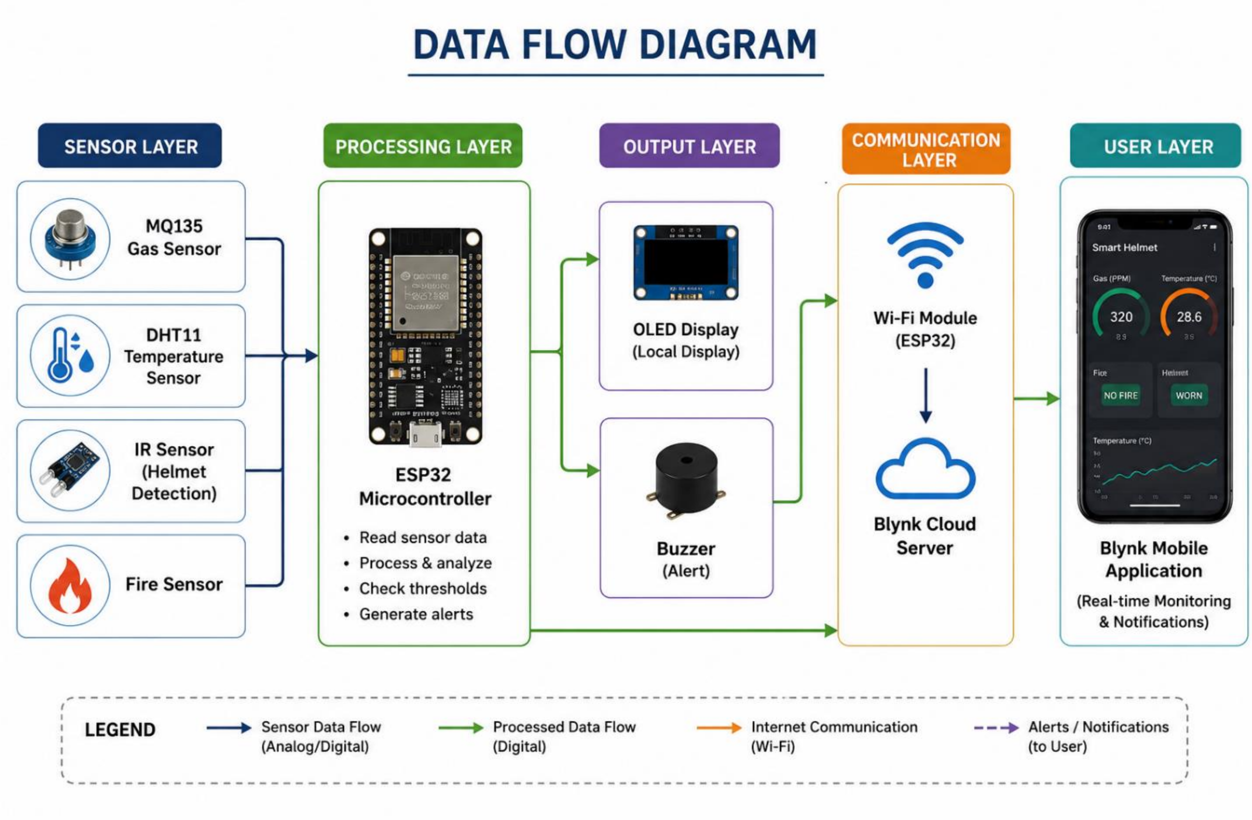


5.5 Interfaces (if applicable)

Hardware Interface

- ESP32 Microcontroller
- MQ135 Gas Sensor
- DHT11 Temperature Sensor
- IR Sensor
- Fire Sensor
- OLED Display (I2C)
- Buzzer
- MT3608 Boost Converter
- 3.7V Battery
- **Software Interface**
- Arduino IDE
- Blynk IoT Platform
- ESP32 Wi-Fi Module
- **Communication Protocols**
- **Wi-Fi:** Used to transmit sensor data from ESP32 to the Blynk Cloud.
- **I2C:** Used for communication between the ESP32 and the OLED Display.
- **GPIO:** Used for interfacing digital sensors (IR, Fire, Buzzer).
- **ADC:** Used to read analog values from the MQ135 Gas Sensor.
- **Data Flow**


DATA FLOW DIAGRAM
INPUT LAYER
PROCESSING LAYER
OUTPUT LAYER
CLOUD LAYER
USER LAYER
MQ135
Gas Sensor
(Analog Output)
DHT11
Temperature &
Humidity Sensor
(Digital Output)
IR Sensor
(Helmet Detection)
(Digital Output)
Fire Sensor
(Digital Output)
ESP32
Microcontroller
• Reads all sensor data
• Processes and checks threshold values
• Controls outputs
• Sends data to cloud via Wi-Fi
OLED Display
(Real-time Display)
Buzzer
(Alert)
Wi-Fi Module
(ESP32 Built-in)
Blynk
Cloud Server
Stores & Processes Data
Smart Helmet
Live Monitoring
Gas (PPM)
120
Temperature (°C)
32.6
Humidity (%)
45
Fire
No Fire
Helmet
Worn
Alert
Normal
Blynk Mobile Application
(Real-time Monitoring & Alerts)
LEGEND
Wired Connection (Data/Control)
Wireless Connection (Wi-Fi)
DATA FLOW SUMMARY
Sensors → ESP32 → Outputs (OLED, Buzzer)
→ Wi-Fi → Blynk Cloud → Mobile App
Real-time Monitoring and Instant Alerts
ABBREVIATIONS
PPM – Parts Per Million
°C – Degree Celsius


DATA FLOW DIAGRAM
SENSOR LAYER
PROCESSING LAYER
OUTPUT LAYER
COMMUNICATION LAYER
USER LAYER
MQ135
Gas Sensor
DHT11
Temperature
Sensor
IR Sensor
(Helmet
Detection)
Fire Sensor
ESP32
Microcontroller
• Read sensor data
• Process & analyze
• Check thresholds
• Generate alerts
OLED Display
(Local Display)
Buzzer
(Alert)
Wi-Fi Module
(ESP32)
Blynk Cloud
Server
Smart Helmet
Gas (PPM)
Temperature (°C)
320
28.6
Fire
Helmet
NO FIRE
WORN
Temperature (°C)
Blynk Mobile
Application
(Real-time Monitoring
& Notifications)
LEGEND
Sensor Data Flow
(Analog/Digital)
Processed Data Flow
(Digital)
Internet Communication
(Wi-Fi)
Alerts / Notifications
(to User)

6 PERFORMANCE TEST

The Smart Helmet system was tested by evaluating its ability to monitor hazardous environmental conditions in real time. The project was validated using different operating scenarios to ensure reliable sensor readings, timely alerts, and continuous communication with the Blynk IoT platform.

6.1 Identified Constraints

Constraint	Impact	Solution Implemented
Sensor Accuracy	Incorrect readings may cause false alarms.	Threshold values were calibrated and sensors were tested multiple times.
Wi-Fi Connectivity	Loss of internet affects remote monitoring.	OLED display and buzzer continue to work locally even if Wi-Fi is unavailable.
Power Consumption	Continuous monitoring drains the battery.	ESP32 and sensors were selected for low-power operation.
Processing Speed	Delay in processing can slow emergency alerts.	ESP32 processes sensor data every 2 seconds, providing near real-time monitoring.
Environmental Conditions	Dust and moisture may affect sensors.	Sensors were positioned carefully and the helmet design protects the electronics.
System Reliability	Continuous operation is required in mining environments.	Multiple sensors provide reliable monitoring of different hazards.

6.2 Test Results

Test Case	Expected Result	Actual Result	Status
Gas Detection	Alert generated when gas exceeds threshold	Alert received on OLED, buzzer, and Blynk	Passed
Temperature Monitoring	Temperature displayed continuously	Accurate readings displayed	Passed

Test Case	Expected Result	Actual Result	Status
Fire Detection	Fire alert generated	Buzzer activated and notification sent	Passed
Helmet Detection	Helmet removal detected	Alert generated successfully	Passed
OLED Display	Live sensor values displayed	Display updated correctly	Passed
Blynk Communication	Sensor data updated on mobile app	Real-time monitoring successful	Passed

6.3 Test Procedure

1. The smart helmet system was powered using a 3.7V battery.
2. The ESP32 was connected to the Wi-Fi network.
3. The system initialized all sensors and established communication with the Blynk IoT platform.
4. Each sensor was tested individually by creating the corresponding environmental condition.
5. The OLED display was monitored to verify real-time sensor readings.
6. The Blynk mobile application was observed for live data updates and alert notifications.
7. The buzzer operation was verified during hazardous conditions.
8. The obtained results were compared with the expected outputs.

6.4 Performance Outcome

7 6.6 Performance Outcome

The Smart Helmet system successfully monitored hazardous environmental conditions in real time. During testing, the ESP32 accurately collected data from the MQ135 gas sensor, DHT11 temperature sensor, IR sensor, and fire sensor. The processed information was displayed on the OLED screen and simultaneously transmitted to the Blynk IoT platform through Wi-Fi.

Whenever abnormal conditions such as gas leakage, fire, high temperature, or helmet removal were detected, the system activated the buzzer and generated instant notifications on the Blynk mobile application. The overall testing confirmed that the system operates reliably and can be effectively used for enhancing miner safety.

Hardware Setup

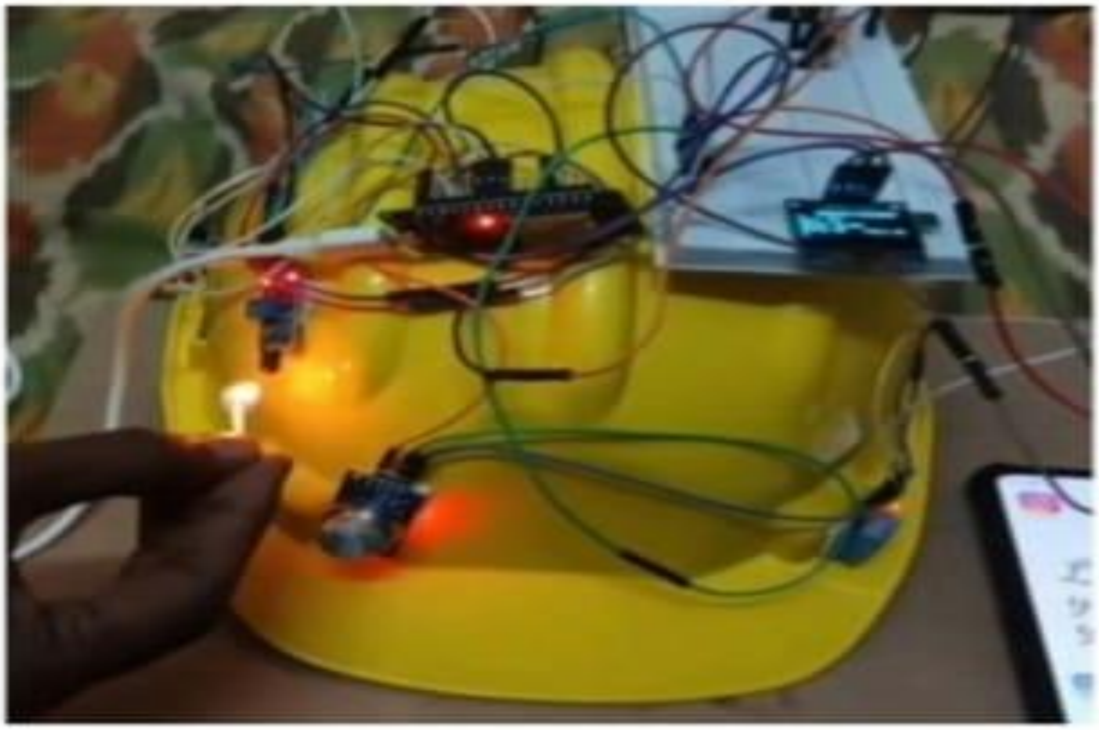


Figure 6.1 Hardware Setup of the Smart Helmet System

OLED Display Output:

Figure 6.2 OLED Display Showing Real-Time Sensor Data

Blynk Notification :

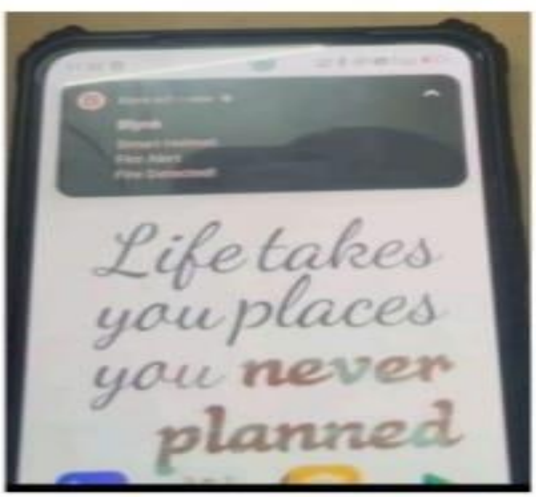


Figure 6.3: Blynk Notification for Hazard Detection

Final Output and Sensor Testing of the Smart Helmet System:

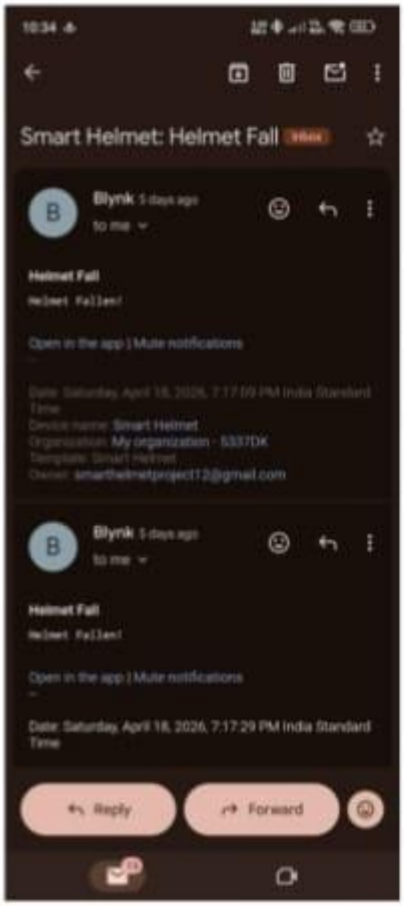
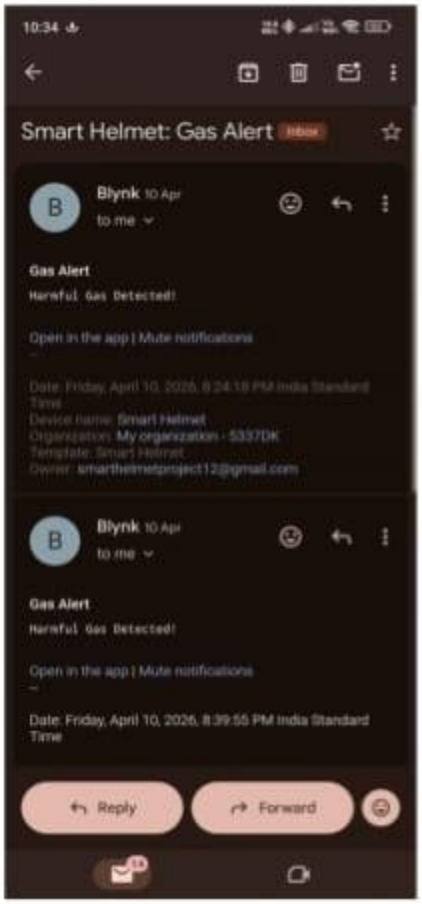


Figure 6.4: Final Output and Sensor Testing of the Smart Helmet System


10:34
Smart Helmet: Gas Alert
Blynk 10 Apr
to me
Gas Alert
Harmful Gas Detected!
Open in the app | Mute notifications
Owner: smarthelmetproject12@gmail.com
Blynk 10 Apr
to me
Gas Alert
Harmful Gas Detected!
Open in the app | Mute notifications
Date: Friday, April 10, 2026, 8:39:55 PM India Standard Time
Reply
Forward

7. MY LEARNINGS

The Embedded Systems & IoT Internship provided me with valuable practical exposure to embedded system design and IoT applications. During the internship, I learned about ESP32 programming, sensor interfacing, real-time data acquisition, Wi-Fi communication, and cloud-based monitoring using the Blynk IoT platform.

I also improved my understanding of project planning, hardware integration, GitHub repository management, technical documentation, and report preparation. Working on this project enhanced my analytical thinking, problem-solving ability, and technical confidence.

Overall, this internship strengthened my knowledge of embedded systems and IoT technologies and motivated me to explore more advanced industrial applications. The skills and experience gained during this internship will be beneficial for my academic projects and future professional career.

8. FUTURE WORK SCOPE

The proposed Smart Helmet system can be further enhanced by integrating GPS technology to provide real-time location tracking of miners during emergency situations. Additional health-monitoring sensors such as heart rate and blood oxygen sensors can also be incorporated to continuously monitor the worker's health.

Future versions of the system may include AI-based predictive analytics for early hazard detection, LoRa or GSM communication for long-range connectivity in underground mines, and improved battery management for extended operation. These enhancements would make the system more intelligent, reliable, and suitable for large-scale industrial deployment.

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